

QBUS2820 Predictive Analytics, S2 2026

Week 4&5: Model Selection (Overfitting & its avoidance)

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Popular model selection methods

Variable selection in linear regression

Recommended reading

- Chapter 6, *An Introduction to Statistical Learning with Applications in R* by James et al.: easy to read, sloppy sometimes, come with R/Python code for practice.
- Chapter 7 *The Elements of Statistical Learning* by Hastie et al.: well-written, deep in theory, suitable for students with a sound maths background.

Introduction

Introduction

Model selection problem

- **Model selection** in general and **variable selection** in particular are important parts of data analysis. Variable selection can be considered as a special case of model selection.
- Model selection is the process of choosing the most appropriate model from **a set of candidate models** for a **given dataset** and task.
- The goal is to select the model that “best” captures the underlying patterns in the data and/or to make good predictions on future observations.
- “best” depends on how we define it: not being overfitting, producing accurate predictions, etc

Introduction

Variable Selection problem

- Consider a regression model with response Y and a set of p potential covariates $X_1, \dots, X_p$.
- At the beginning stage of modelling, p is often large in order to reduce possible bias
 - a large p might cause heavy admin duties, be costly, etc
 - more importantly, a large p typically leads to a high variance in prediction (see later)
- The Variable Selection problem is to select the “best” subset of these p covariates to explain/predict Y .
- “best” depends on how we define it: not being overfitting, producing accurate predictions, etc

Model selection & Variable Selection problem

For example, given a dataset

$D = \{(x_{i1}, x_{i2}, y_1), \dots, (x_{n1}, x_{n2}, y_n)\}$. Five models are proposed to explain D :

Model 1: $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$.

Model 2: $y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$.

Model 3: $y_i = \beta_0 + \beta_1 x_{i2} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2)$.

Model 4: KNN model with $K = 5$.

Model 5: KNN model with $K = 20$.

Then, you need to answer the question: which model is better?

Model Selection problem

- George Box: *All models are wrong, but some are useful!*
- A statistical model is a mathematical simplification of the nature. It cannot reflect exactly the nature which is often a super complicated system
- When we say that model A is better than model B, what do we mean by “better”? We need a (quantitative) way to compare between models.
- This lecture sets out the basic concepts of model selection, and covers methods for model comparison.

Test error & Training error

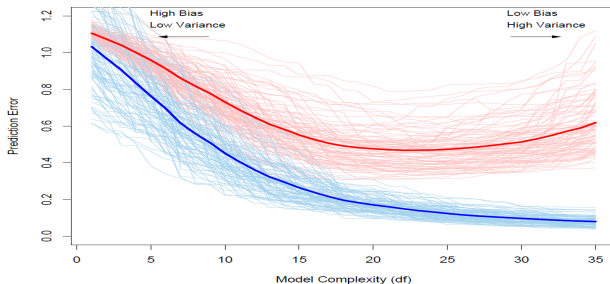
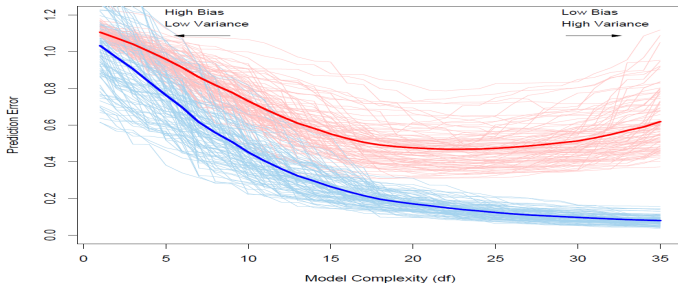


Figure 1: @ESL book

The **light red** curves show the conditional test error $\text{Err}_{\mathcal{T}}$ for 100 training sets of size 50 each. The **solid red** curve shows the expected test error Err (averaged over the conditional test errors). The x -axis is the model complexity - proportional to the number of predictors used in the model

Basic concepts

Training error is not good for model selection



The **light blue** curves show the training errors $\overline{err}$ for 100 training sets of size 50 each. The **solid blue** curve shows the expected training error $\mathbb{E}(\overline{err})$ (averaged over the training errors). Training errors consistently decrease when model complexity increases

Basic concepts

- Suppose that we have a target variable Y , an input vector X and a prediction model $\hat{f}(X)$ that has been estimated from training set $\mathcal{T}$.
- The loss function for measuring errors between Y and $\hat{f}(X)$ is denoted by $L(Y, \hat{f}(X))$, e.g.
 - Squared error loss: $L(Y, \hat{f}(X)) = (Y - \hat{f}(X))^2$
 - 0-1 loss: $L(Y, \hat{f}(X)) = I(Y \neq \hat{f}(X))$
 - Log-likelihood loss: $L(Y, \hat{f}(X)) = -\log p(Y|\hat{f}(X))$

Basic concepts

Model selection criterion: test error (prediction error)

The **test error**, also referred to as *generalization error*, is the prediction error over an independent test sample

$$\text{Err}_{\mathcal{T}} = \mathbb{E}_{(X,Y)}[L(Y, \hat{f}(X)) | \mathcal{T}];$$

where X and Y are drawn randomly from their joint distribution (population).

- Here the training set $\mathcal{T}$ is fixed, and the test error refers to the error for this specific training set.
- Note that $\text{Err}_{\mathcal{T}}$ is dependent on the training data $\mathcal{T}$, and is therefore a **random quantity**.

Basic concepts

Model selection criterion: expected test error (expected prediction error)

A related quantity is the **expected test error** (or expected test error)

$$\text{Err} = \mathbb{E}[L(Y, \hat{f}(X))] = \mathbb{E}_{\mathcal{T}}[\text{Err}_{\mathcal{T}}].$$

- Note that this expectation averages over everything that is random, **including the randomness in the training set that produced $\hat{f}$** .
- This expected prediction error is an ideal measure of performance for model selection.
- In this chapter, we describe a number of methods for **estimating the expected test error** for a model.

Basic concepts

Training error

Training error is the average loss over the training sample,

$$\overline{\text{err}} = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(x_i))$$

- Given a model, training data is used to fit the model (i.e. estimating the parameters), e.g. by minimising the training error.
- Training error is not a good measure of model selection, it decreases when model complexity increases
- Check yourself: think of k in kNN, and p in linear regression.

Training, validation and test set

- If we are in a data-rich situation, the best approach for both problems is to randomly divide the dataset into three parts: a training set, a validation set, and a test set.
- Training data: used to fit the models;
- Validation data: used to estimate prediction error and select the optimal model;
- Test data: to check the performance of the selected model.

Note that only '**training data**' and '**validation data**' are used in building a model. Test data presents a real performance check when the model is used in practice.

To be continued...

- The methods in this chapter are designed for situations where there is insufficient data to split it into three parts.
- The methods of this chapter approximate the validation step either **analytically (AIC, BIC)** or by **efficient sample re-use (cross-validation)**.

Bias and Variance Decomposition

Consider the general regression model

$$Y = f(X) + \varepsilon, \quad \mathbb{E}(\varepsilon) = 0, \quad \mathbb{V}(\varepsilon) = \sigma^2$$

Let $\hat{f}(X)$ be an estimate of $f(X)$. Using squared-error loss, the expected prediction error at x_0 is

$$\begin{aligned}\text{Err}(x_0) &= \mathbb{E}\left[(Y - \hat{f}(x_0))^2 | X = x_0\right] \\ &= \mathbb{E}\left[(\hat{f}(x_0) - f(x_0))^2\right] + \sigma^2 \\ &= \mathbb{E}\left[\left((\hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)]) + (\mathbb{E}[\hat{f}(x_0)] - f(x_0))\right)^2\right] + \sigma^2 \\ &= \mathbb{V}(\hat{f}(x_0)) + \left(\mathbb{E}[\hat{f}(x_0)] - f(x_0)\right)^2 + \sigma^2 \\ &= \text{Variance}(\hat{f}) + \text{Bias}^2(\hat{f}) + \sigma^2\end{aligned}$$

As σ^2 is the irreducible error, the EPE is basically decomposed into two terms: Variance and Bias².

Example: KNN

Recall that

$$\hat{f}(x_0) = \frac{1}{k} \sum_{x_i \in N_k(x, \mathcal{D})} y_i,$$

where $N_k(x, \mathcal{D})$ is the neighbourhood set containing k input points among $\{x_i\}_{i=1}^N$ that are closest to x_0 .

- For simplicity, we assume that training inputs x_i are fixed, and the randomness arises from the y_i .

Then

$$\begin{aligned} \text{Bias}(\hat{f}) &= \mathbb{E}[\hat{f}(x_0)] - f(x_0) \\ &= \mathbb{E}\left[\frac{1}{k} \sum_{x_i \in N_k(x, \mathcal{D})} (f(x_i) + \varepsilon_i)\right] - f(x_0) \\ &= \frac{1}{k} \sum_{x_i \in N_k(x, \mathcal{D})} f(x_i) - f(x_0) \end{aligned}$$

Hence, the Bias² increases as k grows.

Example: KNN

For the k-nearest-neighbor regression fit,

$$\begin{aligned}\text{Variance}(\hat{f}) &= \mathbb{V}\left(\hat{f}(x_0)\right) \\ &= \mathbb{V}\left(\frac{1}{k} \sum_{x_i \in N_k(x, \mathcal{D})} \varepsilon_i\right) \\ &= \frac{1}{k^2} \mathbb{V}\left(\sum_{x_i \in N_k(x, \mathcal{D})} \varepsilon_i\right) \\ &= \frac{1}{k^2} \sum_{x_i \in N_k(x, \mathcal{D})} \mathbb{V}(\varepsilon_i) \\ &= \frac{\sigma^2}{k}\end{aligned}$$

Hence, the Variance decreases as k grows.

Example: Linear regression*

For a linear model fit $\hat{f}_p(x) = x'\hat{\beta}$, where the parameter vector β with p components is fit by least squares, we have

$$\begin{aligned}\text{Bias}^2 &= (f(x_0) - \mathbb{E}[\hat{f}_p(x_0)])^2 \\ &= [f(x_0) - x_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'f(\mathbf{X})]^2,\end{aligned}$$

$$\begin{aligned}\text{Variance}(\hat{f}) &= \mathbb{V}(x_0'\hat{\beta}) = x_0'\mathbb{V}(\hat{\beta})x_0 \\ &= x_0'(\mathbf{X}'\mathbf{X})^{-1}x_0 \cdot \sigma^2 \\ &= \sigma^2 \sum_{i=1}^p x_{0i}^2,\end{aligned}$$

Here, we've supposed that the $n \times p$ -dimensional design matrix $\mathbf{X}$ is such that $\mathbb{V}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2 I$.

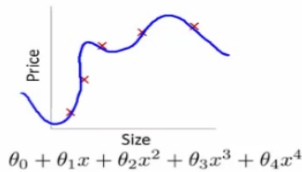
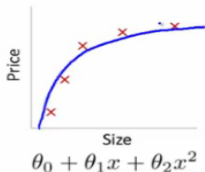
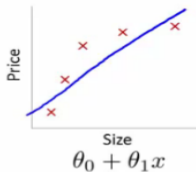
So, the more number of covariates p is, the bigger the variance and the smaller the bias², and vice versa.

Basic concepts

Overfitting

$$\text{Expected prediction error} = \text{Variance} + \text{Bias}^2$$

- Overfitting: A complex model is used ($\hat{f}(x)$ is a complicated function, involves a lot of parameters, etc) then Bias^2 is small but Variance is large. Underfitting: otherwise.
- Model selection: pick up a right model that trades off between Bias and Variance



Review questions

- What is the model selection problem?
- What is the variable selection problem?
- What is the difference between prediction error and expected prediction error?
- How does model selection relate to the bias-variance trade-off?
- What is a validation set? How is it different from a test set?

Popular model selection methods

Recall...

- **Model selection** in general and **variable selection** in particular are important parts of data analysis.
- Let $\{M_i, i \in I\}$ be a set of potential models, we want to select the best one.
- Variable selection is a special case of model selection: select the “best” subset of the given p predictors to explain/predict the response Y .
- Two ideal measures for doing model selection: prediction error and expected prediction error
- Bias-Variance decomposition

$$\text{Expected prediction error} = \text{Variance} + \text{Bias}^2$$

So model selection is to pick up a right model that trades off between Bias and Variance

Basic concepts

Penalised maximum likelihood principle

- Typically, the training error $\overline{err}$ is smaller than the prediction error, because the same data is used to fit the model and assess its error (see the picture).
- Think of what happens if the player is also the referee! This is similar to using training error to judge model fitting.
- As training error decreases as model complexity increases, it might be a good idea to penalise for model complexity
- Many popular model selection criteria have the form
model selection criterion = $\overline{err}$ + penalty of model complexity

Akaike's information criterion (AIC): Select the model with the smallest AIC

$$\text{AIC} = -2 \times \log\text{-likelihood}(\hat{\theta}_{mle}) + 2d$$

which has the form: training error+model complexity.

- $\log\text{-likelihood}(\hat{\theta}_{mle})$ is the log-likelihood evaluated at the MLE $\hat{\theta}_{mle}$.
- d is the number of parameters in θ (i.e. # covariates).
- The factor 2 is not important, but useful when comparing AIC to other model selection criteria.
- proposed by Hirotugu Akaike in 1973.

AIC: theoretical justification*

Let $f_d(x; \theta)$ be a family of likelihood with the complexity indexed by d . Suppose there are both training data and validation data available.

- estimate θ by maximizing $\sum_{x \in \mathcal{T}} \ln(f_d(x; \theta))$, leading to $\hat{\theta}_{d,mle}$, where the sum is taken over all x in the training data set.
- choosing d to maximize $\sum_{x \in \mathcal{V}} \ln(f_d(x; \hat{\theta}_{d,mle}))$, where the sum is taken over all x in the validation data.

In practice, we only have one data set, $-2 \sum_{x \in \mathcal{V}} \ln(f_d(x; \hat{\theta}_{d,mle}))$ has the same asymptotic mean as

$$\text{AIC} = -2 \times \log\text{-likelihood}(\hat{\theta}_{d,mle}) + 2d.$$

Bayesian information criterion (BIC): Select the model with the smallest BIC

$$\text{BIC} = -2 \times \log\text{-likelihood}(\hat{\theta}_{mle}) + (\log n) \times d$$

BIC, proposed by Gideon Schwarz 1978, is motivated by Bayesian approach.

BIC: theoretical justification*

Consider a model $\mathcal{M}$ with d -vector of parameters θ , and training data $\mathcal{T}$. The posterior of $\mathcal{M}$ is

$$\begin{aligned} p(\mathcal{M}|\mathcal{T}) &\propto p(\mathcal{M})p(\mathcal{T}|\mathcal{M}) \\ &= p(\mathcal{M}) \int p(\mathcal{T}|\theta, \mathcal{M})p(\theta|\mathcal{M})d\theta. \end{aligned}$$

Here: $p(\mathcal{T}|\theta, \mathcal{M})$ is the likelihood function (under model $\mathcal{M}$) and $p(\theta|\mathcal{M})$ is the prior distribution of θ .

- Bayesian statistics encodes all information about model $\mathcal{M}$ in its posterior distribution $p(\mathcal{M}|\mathcal{T})$.
- Bayesian statistics is getting more and more popular in modern statistics. Taught in QBUS3830.
- We want a model $\mathcal{M}$ with the highest posterior $p(\mathcal{M}|\mathcal{T})$, which is equivalent to the highest $p(\mathcal{T}|\mathcal{M})$ under certain assumptions.

BIC: theoretical justification*

Using an uniform prior for $\mathcal{M}$ and approximating the integral by the so-called Laplace approximation, it can be shown that

$$\begin{aligned} -2 \log p(\mathcal{T}|\mathcal{M}) &\approx -2 \log p(\mathcal{T}|\hat{\theta}_{mle}, \mathcal{M}) + d \log(n) \\ &= \text{BIC}. \end{aligned}$$

We want to pick a model with the highest posterior $p(\mathcal{M}|\mathcal{T})$, which is equivalent to picking a model with the smallest BIC.

AIC or BIC?

$$\text{AIC} = -2 \times \log\text{-likelihood}(\hat{\theta}_{mle}) + 2d$$

$$\text{BIC} = -2 \times \log\text{-likelihood}(\hat{\theta}_{mle}) + (\log n) \times d$$

- They're both popular model selection methods. BIC puts a heavier penalty on model complexity, thus often leading to a simpler model.
- AIC aims to find the **most predictive** model.
- BIC assumes that one of the models is true and that you are trying to find the **model most likely to be true** in the Bayesian sense.
- AIC tends to overestimate the number of predictors, while BIC gives a consistent estimator for the true d : it is able to identify the true model when $n \rightarrow \infty$ (if there exists such a true model)
- On the other hand, for finite samples, BIC often chooses models that are too simple, because of its heavy penalty on complexity.

AIC and BIC in Gaussian linear regression

Recall linear regression

$$\mathbf{y} = \mathbf{X}\beta + \epsilon, \quad \epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2).$$

Then AIC and BIC can be conveniently defined as

$$\text{AIC} = n \times \log(\text{RSS}/n) + 2d$$

$$\text{BIC} = n \times \log(\text{RSS}/n) + (\log n) \times d,$$

where $\text{RSS} = \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 = (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta})$.

- Note that if all linear models have the **same d** , minimizing AIC or BIC is **equivalent to minimizing RSS**.

Cross-validation (CV)

- basic idea: like you validate your peers' work and they validate yours
- probably the simplest but most commonly used model selection method
- give an estimate of the expected prediction error

Cross-validation

- divide the data into K sets, $K \geq 2$. Often, this is done randomly
- for the k th part, fit the model to the other $K - 1$ parts. Denote the fitted model as $\hat{f}^{-k}(x)$
- use the fitted model to predict the k th part. The prediction error is

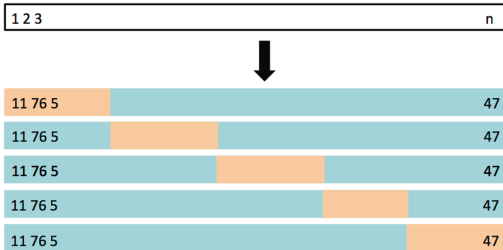
$$\sum_{(y_i, x_i) \in \text{part } k} L(y_i, \hat{f}^{-k}(x_i))$$

- The k -fold cross-validated prediction error is

$$CV = \frac{1}{n} \sum_{k=1}^K \sum_{(y_i, x_i) \in \text{part } k} L(y_i, \hat{f}^{-k}(x_i))$$

It's an estimate of the expected prediction error as it's averaged over both test data and training data

K -fold CV



Cross-validation

- The selected model is the one with the smallest CV prediction error.
- Typical choices of K are 5, 10 or n . The case $K = n$ is known as **leave-one-out** cross-validation.
- Refit on the entire dataset after cross-validation.

Cross-validation is simple and widely used. However, CV can be sometimes very computationally expensive because one has to fit the model many times.

Variable selection in linear regression

Variable selection in linear regression

- Consider a linear regression model with p potential covariates.
- At the initial step of modelling, a large number p of covariates is often introduced in order to reduce potential bias.
- We then aim to identify a subset of p predictors that we believe to be related to the response.
- Finally we fit a model using least squares on the selected set of variables.

Best subset selection: Search over the totally 2^p possible subsets of p covariates to find the best subset. The criterion can be AIC, BIC or any other model selection criteria.

Best subset selection

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \dots, p$
 - a Fit all $\binom{p}{k}$ models that contain exactly k predictors.
 - b Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$.
Here *best* is defined as having the smallest *residual sum of squares* (RSS) or equivalently largest R^2 .
3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using cross-validated prediction error, AIC, BIC, or adjusted R^2 .

Variable selection in linear regression

Searching over 2^p subsets is only feasible when p is small (< 30)

Forward-stepwise selection: Start with the intercept, then sequentially add into the model the covariate that most improves the model selection criterion.

Backward-stepwise selection: Start with the full model with p covariates, then sequentially remove the covariate whose removal most improves the model selection criterion.

- Advantage: much more time-efficient than the best subset selection method
- Disadvantage: not necessarily end up at the best subset.

Forward stepwise selection (FSS)

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 1, 2, \dots, p - 1$:
 - a Fit all $p - k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
 - b Pick the *best* among these $p - k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having the smallest RSS or equivalently largest R^2 .
3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using cross-validated prediction error, AIC, BIC, or adjusted R^2 .

Pros and cons of FSS

- Computational advantageous to the best subset selection approach
- FSS approach searches through $1 + p(p + 1)/2$ models.
- It is not guaranteed to find the best possible model out of all 2^p models containing subsets of the p predictors. Why? Any example?

Credit data example

# Variables	Best subset	Forward stepwise
One	rating	rating
Two	rating, income	rating, income
Three	rating, income, student	rating, income, student
Four	cards, income student, limit	rating, income, student, limit

The first four selected models for best subset selection and forward stepwise selection on the Credit data set. The first three models are identical but the fourth models differ.

Backward stepwise selection (BSS)

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
2. For $k = p, p - 1, \dots, 1$:
 - a Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k - 1$ predictors.
 - b Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having the smallest RSS or equivalently largest R^2 .
3. Select a single best model from among $\mathcal{M}_0, \dots, \mathcal{M}_p$ using cross-validated prediction error, AIC, BIC, or adjusted R^2 .

More about BSS

- Like FSS, BSS approach searches through $1 + p(p + 1)/2$ models and so can be applied in settings where p is too large to apply best subset selection.
- Like FSS, BSS is not guaranteed to yield the best model containing a subset of p predictors.
- BSS requires the number of observations n be larger than p so that the full model can be fit, however FSS can be used even when $n < p$ and so is the only viable subset method when p is very large.

Choosing the optimal Model

- The model containing more predictors tend to produce smaller RSS and larger R^2 . Why?
- We wish to choose the model with low expected prediction error/test error.
- Therefore, RSS and R^2 are not appropriate to be set as a criterion for selecting the best model among a collection of models with different number of predictors.
 1. We can estimate the expected prediction error by **making adjustment to the training error** to account for the bias due to overfitting.
 2. Or a different **cross validation** type of approaches can be applied.

Variable selection in linear regression

Variable selection based on hypothesis testing

- Consider the test $H_0 : \beta_j = 0$ v.s. $H_1 : \beta_j \neq 0$.
- Let $\hat{\beta}_j$ be an estimator of β_j . If the sampling distribution of $\hat{\beta}_j$ is known, p -value can be computed
- If the p -value is large (e.g. $> 0.05, 0.1$) then the corresponding covariate X_j might be removed from the model

Possible disadvantages

- It's not clear what prediction error is optimised
- not time-efficient when p is large (need to refit the model many times)

This variable selection method is not popular in “modern” statistics, machine learning. Because of historic reasons, it's still widely used in many fields such as social sciences.

Woman labor force example

The data set “MROZ.xlsx”, available in Blackboard, contains information on Women’s labour force participation.

We would like to build a logistic regression model to explain women’s labour force participation using potential predictors `nwifeinc` (income), `educ` (years of education), `age`, `exper` (years of experience), `expersq` (squared years of experience), `kidslt6` (number of kids less than 6-year-old) and `kidsge6` (number of kids more than 6-year-old).

Let’s carry out the variable selection task.

Woman labor force example

Generalized Linear regression model:

```
logit(inlf) ~ nwifeinc + educ + exper + exper2 + age + age2 + kidslt6 + kidsge6  
Distribution = Binomial
```

Estimated Coefficients:

	Estimate	SE	tStat	pValue
nwifeinc	-0.021384	0.0084038	-2.5446	0.010941
educ	0.21817	0.042536	5.1292	2.9101e-07
exper	0.20393	0.032365	6.301	2.9568e-10
exper2	-0.0030799	0.0010308	-2.9879	0.0028087
age	-0.063422	0.030944	-2.0496	0.040405
age2	-0.00030134	0.00046495	-0.6481	0.51692
kidslt6	-1.4414	0.19855	-7.2598	3.8765e-13
kidsge6	0.052348	0.07652	0.68411	0.49391



Woman labor force example

Generalized Linear regression model:

$\text{logit}(\text{inlf}) \sim \text{nwifeinc} + \text{educ} + \text{exper} + \text{exper2} + \text{age} + \text{kidslt6} + \text{kidsge6}$

Distribution = Binomial

Estimated Coefficients:

	Estimate	SE	tStat	pValue
nwifeinc	-0.021775	0.0083797	-2.5985	0.0093635
educ	0.23398	0.035063	6.6731	2.5046e-11
exper	0.20829	0.031682	6.5744	4.885e-11
exper2	-0.0032452	0.0010001	-3.2448	0.0011752
age	-0.082575	0.0094592	-8.7296	2.555e-18
kidslt6	-1.4123	0.19317	-7.3114	2.6442e-13
kidsge6	0.077378	0.066154	1.1697	0.24213

Woman labor force example

Generalized Linear regression model:

```
logit(inlf) ~ nwifeinc + educ + exper + exper2 + age + kidslt6
```

```
Distribution = Binomial
```

Estimated Coefficients:

	Estimate	SE	tStat	pValue
nwifeinc	-0.021781	0.00841	-2.5899	0.0095998
educ	0.24405	0.034171	7.1421	9.1946e-13
exper	0.20768	0.031581	6.5762	4.8264e-11
exper2	-0.0033395	0.00099609	-3.3526	0.00080062
age	-0.082574	0.0094581	-8.7305	2.5356e-18
kidslt6	-1.4129	0.19416	-7.2771	3.4099e-13

Review questions

- How can you use the AIC method?
- How can you use the BIC method?
- What is the cross-validation method?
- How would you do variable selection in linear regression?